

Application Deployment with CI/CD using AWS, Terraform, Jenkins and Docker

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Project: Trend-DevOps-Project

Objective

Deploy a web application using an end-to-end CI/CD pipeline with Terraform, Jenkins, Docker and AWS EC2.

Tech Stack

- AWS EC2
- Terraform
- Jenkins
- Docker
- GitHub
- Nginx
- Linux

Architecture

GitHub Repository



Jenkins Pipeline



Docker Build



Docker Container



AWS EC2 Instance



Trendify Application

Implementation Steps

1. Created infrastructure using Terraform.
2. Provisioned EC2 instance and configured security groups.
3. Installed Jenkins and Docker.
4. Integrated GitHub repository with Jenkins.
5. Created Jenkinsfile for CI/CD pipeline.

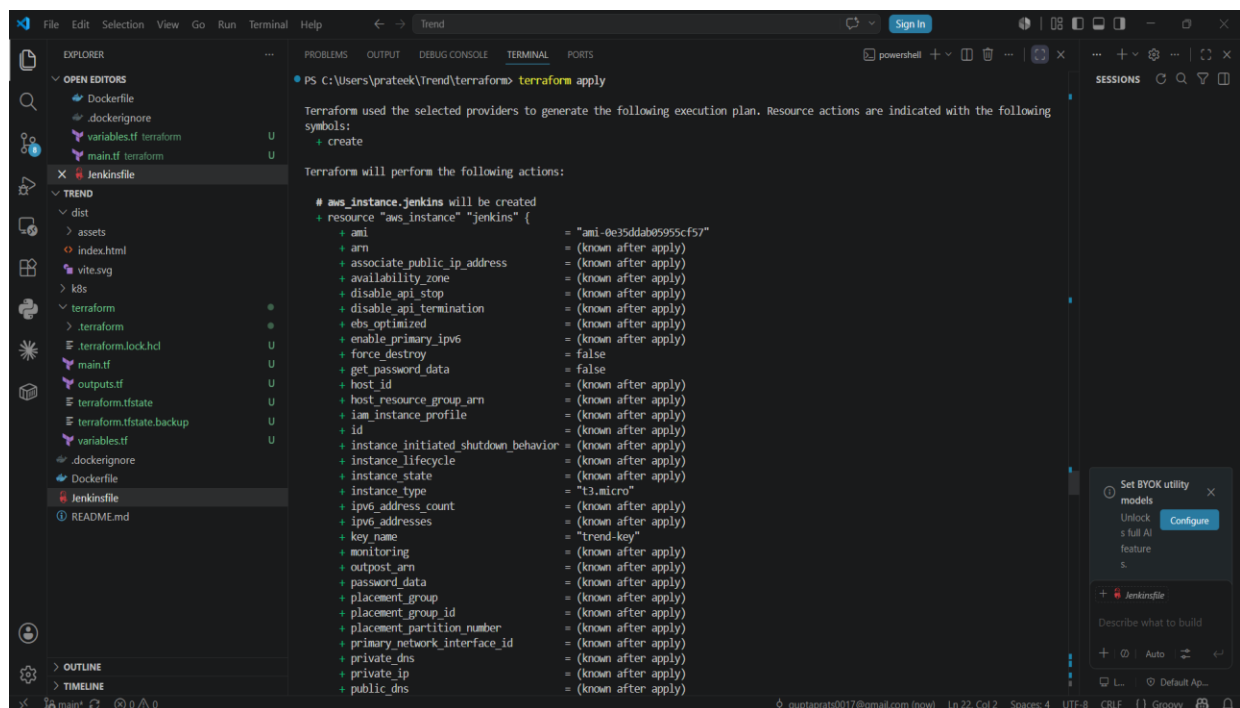
6. Built Docker image automatically.
7. Ran Docker container exposing port 8081.
8. Configured AWS Security Group for ports 22, 8080 and 8081.
9. Verified application deployment.

Results

- ✓ Jenkins pipeline executed successfully.
 - ✓ Docker image built successfully.
 - ✓ Container deployed successfully.
 - ✓ Trendify application accessible externally.
- URL: <http://13.201.62.188:8081>

Screenshots to Attach

1. Terraform apply output



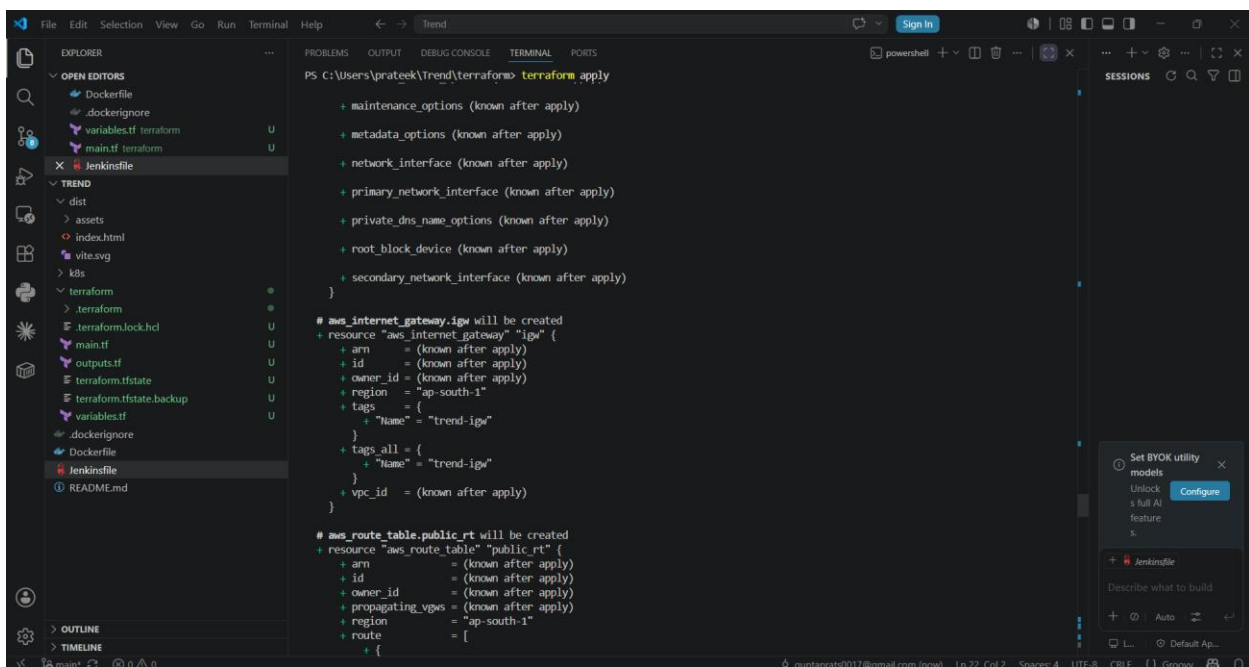
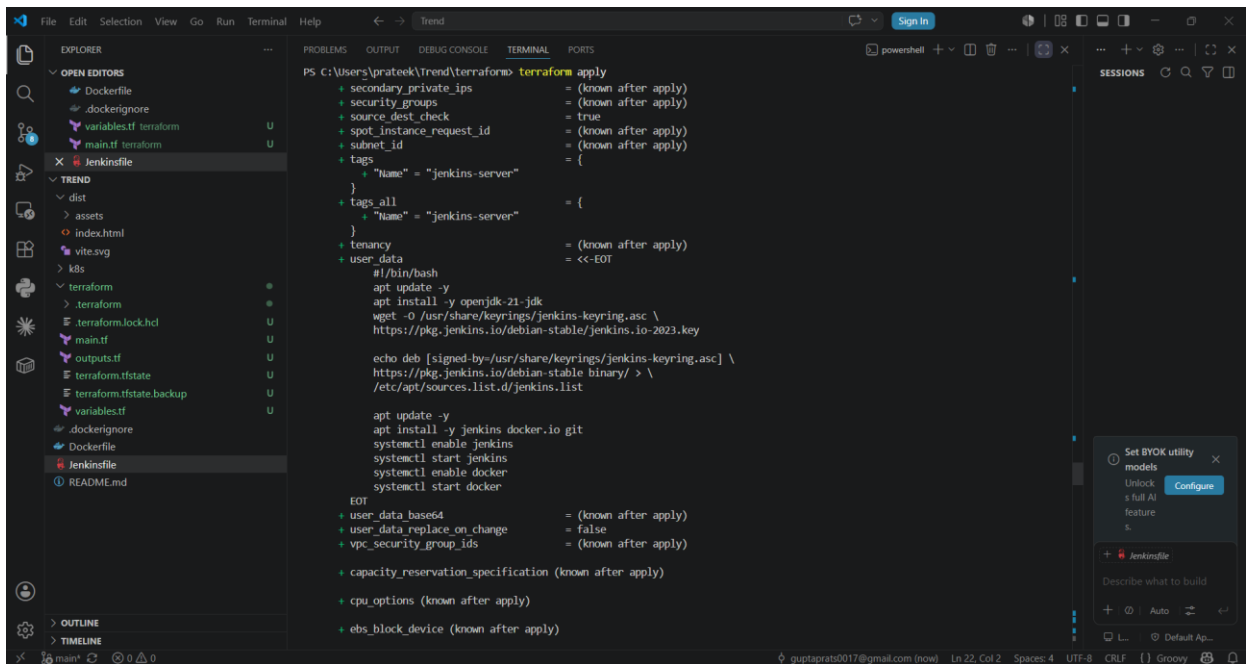
The screenshot shows the Visual Studio Code interface with the Terraform apply output displayed in the terminal. The Explorer pane on the left shows the project structure, including files like Dockerfile, .dockerignore, variables.tf, main.tf, Jenkinsfile, and README.md. The terminal output shows the Terraform execution plan and the actions it will perform.

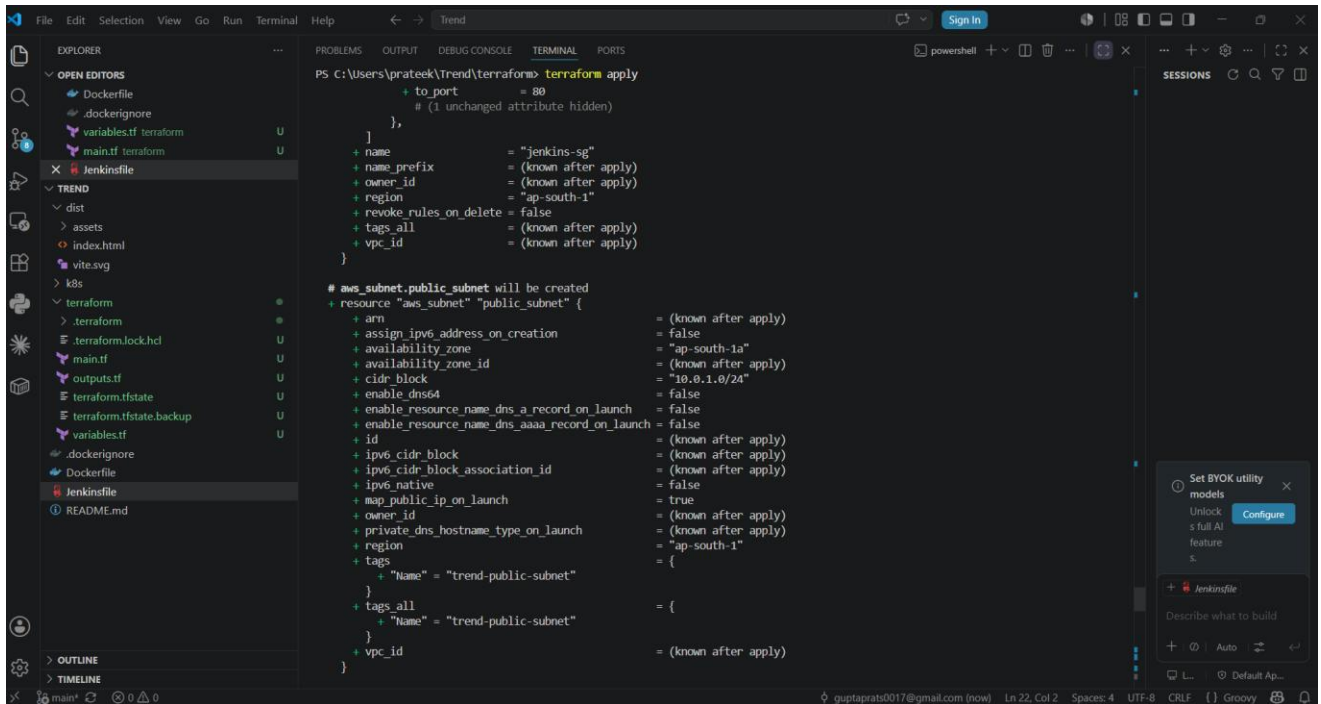
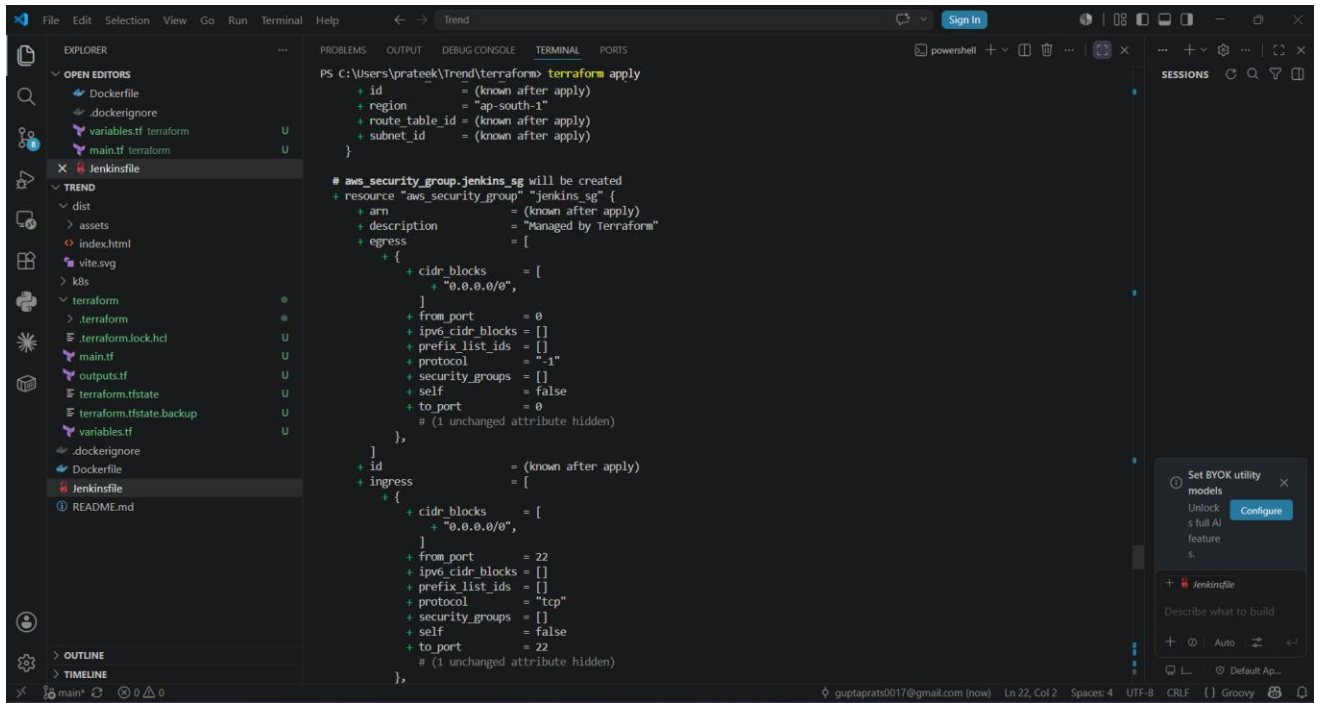
```
PS C:\Users\prateek\trend\terraform> terraform apply

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following
symbols:
+ create

Terraform will perform the following actions:

# aws_instance.jenkins will be created
+ resource "aws_instance" "jenkins" {
  + ami                    = "ami-0e35ddab09955cf57"
  + arn                   = (known after apply)
  + associate_public_ip_address = (known after apply)
  + availability_zone      = (known after apply)
  + disable_api_stop       = (known after apply)
  + disable_api_termination = (known after apply)
  + ebs_optimized          = (known after apply)
  + enable_primary_ipv6    = (known after apply)
  + force_destroy          = false
  + get_password_data      = false
  + host_id                = (known after apply)
  + host_resource_group_arn = (known after apply)
  + iam_instance_profile    = (known after apply)
  + id                     = (known after apply)
  + instance_initiated_shutdown_behavior = (known after apply)
  + instance_lifecycle     = (known after apply)
  + instance_state         = (known after apply)
  + instance_type          = "t3.micro"
  + ipv6_address_count     = (known after apply)
  + ipv6_addresses        = (known after apply)
  + key_name               = "trend-key"
  + monitoring             = (known after apply)
  + outpost_arn            = (known after apply)
  + password_data          = (known after apply)
  + placement_group        = (known after apply)
  + placement_group_id     = (known after apply)
  + placement_partition_number = (known after apply)
  + primary_network_interface_id = (known after apply)
  + private_dns            = (known after apply)
  + private_ip             = (known after apply)
  + public_dns             = (known after apply)
```





```
PS C:\Users\prateek\Trend\terraform> terraform apply

# aws_vpc.trend_vpc will be created
+ resource "aws_vpc" "trend_vpc" {
  + arn                               = (known after apply)
  + cidr_block                        = "10.0.0.0/16"
  + default_network_acl_id           = (known after apply)
  + default_route_table_id           = (known after apply)
  + default_security_group_id        = (known after apply)
  + dhcp_options_id                  = (known after apply)
  + enable_dns_hostnames              = (known after apply)
  + enable_dns_support                = true
  + enable_network_address_usage_metrics = (known after apply)
  + id                               = (known after apply)
  + instance_tenancy                  = "default"
  + ipv6_association_id               = (known after apply)
  + ipv6_cidr_block                   = (known after apply)
  + ipv6_cidr_block_network_border_group = (known after apply)
  + main_route_table_id               = (known after apply)
  + owner_id                         = (known after apply)
  + region                           = "ap-south-1"
  + tags                             = {
    + "Name" = "trend-vpc"
  }
  + tags_all                         = {
    + "Name" = "trend-vpc"
  }
}

Plan: 7 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ instance_public_dns = (known after apply)
+ instance_public_ip  = (known after apply)

Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes
```

```
PS C:\Users\prateek\Trend\terraform> terraform apply

+ tags_all = {
  + "Name" = "trend-vpc"
}

Plan: 7 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ instance_public_dns = (known after apply)
+ instance_public_ip  = (known after apply)

Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

aws_vpc.trend_vpc: Creating...
aws_vpc.trend_vpc: Creation complete after 1s [id=vpc-0a1fada252cf700fb]
aws_internet_gateway.igw: Creating...
aws_subnet.public_subnet: Creating...
aws_security_group.jenkins_sg: Creating...
aws_internet_gateway.igw: Creation complete after 1s [id=igw-0d8626651f9921062]
aws_route_table.public_rt: Creating...
aws_route_table.public_rt: Creation complete after 1s [id=rtb-012b2834942be227a]
aws_security_group.jenkins_sg: Creation complete after 3s [id=sg-0be5c5ab2026bcf0e]
aws_subnet.public_subnet: Still creating... [00m10s elapsed]
aws_subnet.public_subnet: Creation complete after 11s [id=subnet-0b230b18d4e53a00b]
aws_route_table_association.rta: Creating...
aws_route_table_association.rta: Creation complete after 1s [id=rta-05d8c5e532efb4600]
aws_instance.jenkins: Still creating... [00m10s elapsed]
aws_instance.jenkins: Creation complete after 13s [id=i-0fd4349e32aebf90d]

Apply complete! Resources: 7 added, 0 changed, 0 destroyed.

Outputs:
instance_public_dns = ""
instance_public_ip  = "13.201.62.188"
```

2. EC2 instance details

The screenshot displays the AWS Management Console interface for an EC2 instance. The left sidebar shows the navigation menu with categories like EC2, Images, and Elastic Block Store. The main content area is titled 'Instance summary for i-0fd4349e32aebf90d (jenkins-server)'. It provides a comprehensive overview of the instance's configuration, including its ID, IP addresses, state, hostname, and various identifiers like VPC ID and Subnet ID. The instance is currently in a 'Running' state.

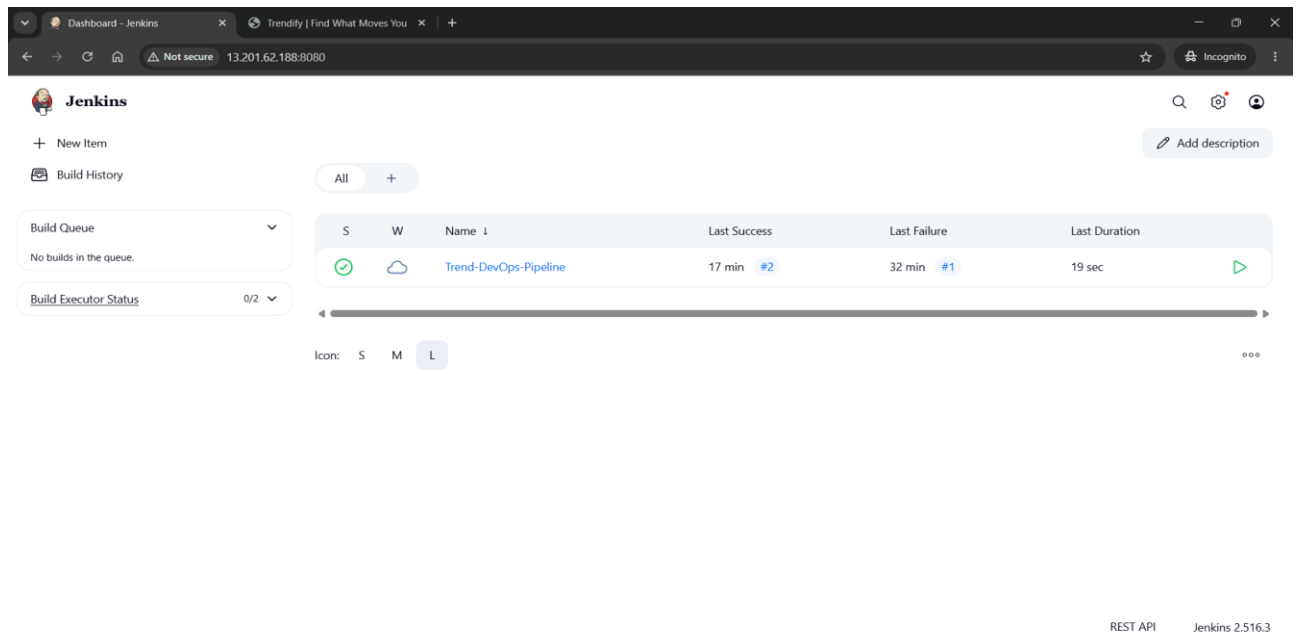
Property	Value
Instance ID	i-0fd4349e32aebf90d
Public IPv4 address	13.201.62.188 open address
Private IPv4 addresses	10.0.1.4
Instance state	Running
Private IP DNS name (IPv4 only)	ip-10-0-1-4.ap-south-1.compute.internal
Instance type	t3.micro
VPC ID	vpc-0a1fada252cf700fb (trend-vpc) link
Subnet ID	subnet-0b230b18d4e53a00b (trend-public-subnet) link
Instance ARN	arn:aws:ec2:ap-south-1:360357042190:instance/i-0fd4349e32aebf90d
IAM role	-
IMDSv2	Required
Auto-assigned IP address	13.201.62.188 [Public IP]
Hostname type	IP name: ip-10-0-1-4.ap-south-1.compute.internal
Answer private resource DNS name	-
Elastic IP addresses	-
AWS Compute Optimizer finding	Opt-in to AWS Compute Optimizer for recommendation s. Learn more
Auto Scaling Group name	-
Managed	false

3. Security Group inbound rules

The screenshot shows the 'Security details' section of the AWS Management Console for the same EC2 instance. It focuses on the 'Inbound rules' of the associated security group, 'sg-0be5c5ab2026bcf08 (jenkins-sg)'. A table lists four inbound rules, each allowing traffic from 0.0.0.0/0 on specific ports (8081, 22, 8080, and 80) using the TCP protocol. The rules are all associated with the 'jenkins-sg' security group.

Name	Security group rule ID	Port range	Protocol	Source	Security groups
-	sgr-00c5320047c3f7c15	8081	TCP	0.0.0.0/0	jenkins-sg link
-	sgr-0021a7b188a637390	22	TCP	0.0.0.0/0	jenkins-sg link
-	sgr-0603d816e5bcd0e45	8080	TCP	0.0.0.0/0	jenkins-sg link
-	sgr-05f225cabb5941b6b	80	TCP	0.0.0.0/0	jenkins-sg link

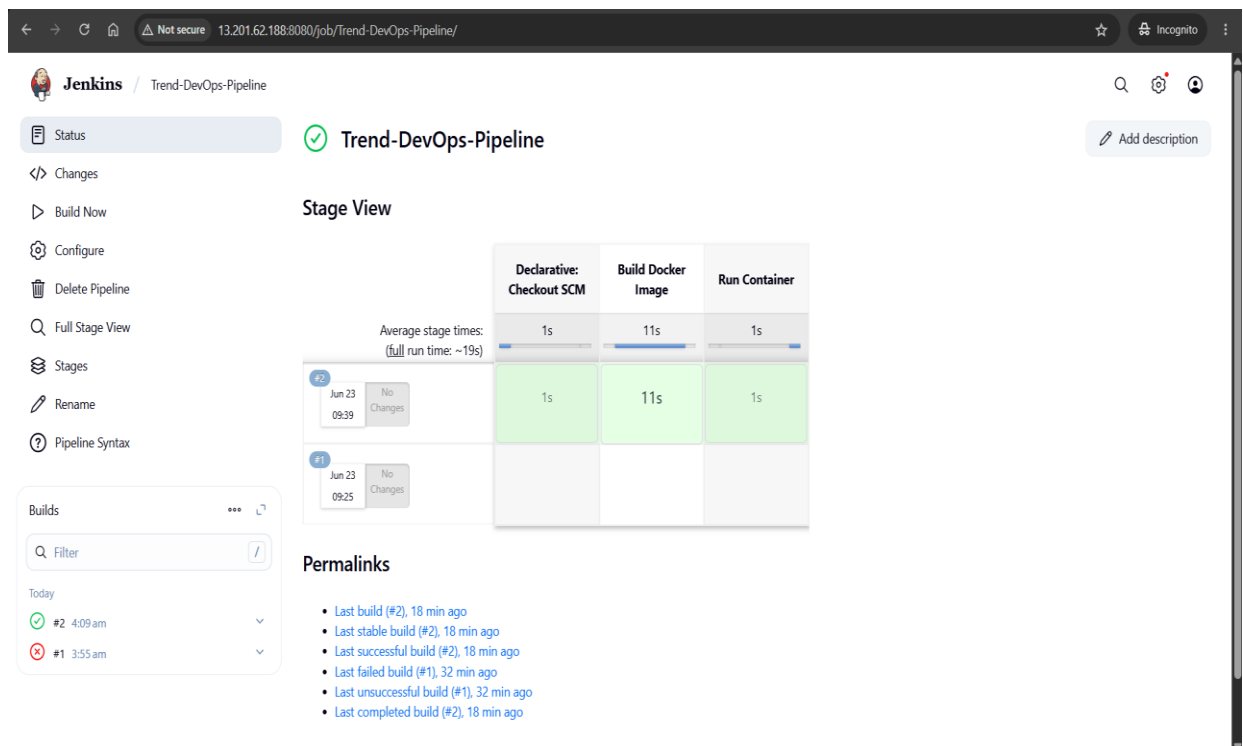
4. Jenkins Dashboard



The screenshot shows the Jenkins Dashboard for a job named 'Trend-DevOps-Pipeline'. The interface includes a sidebar with 'New Item' and 'Build History' links. The main area displays a table of build history with columns for status, name, last success, last failure, and last duration. The table shows two builds: #1 (failed) and #2 (successful). The 'Build Queue' section indicates no builds are in the queue. The 'Build Executor Status' shows 0/2 executors. The bottom right corner displays 'REST API' and 'Jenkins 2.516.3'.

S	W	Name	Last Success	Last Failure	Last Duration
✓	☁	Trend-DevOps-Pipeline	17 min #2	32 min #1	19 sec

5. Jenkins pipeline successful execution



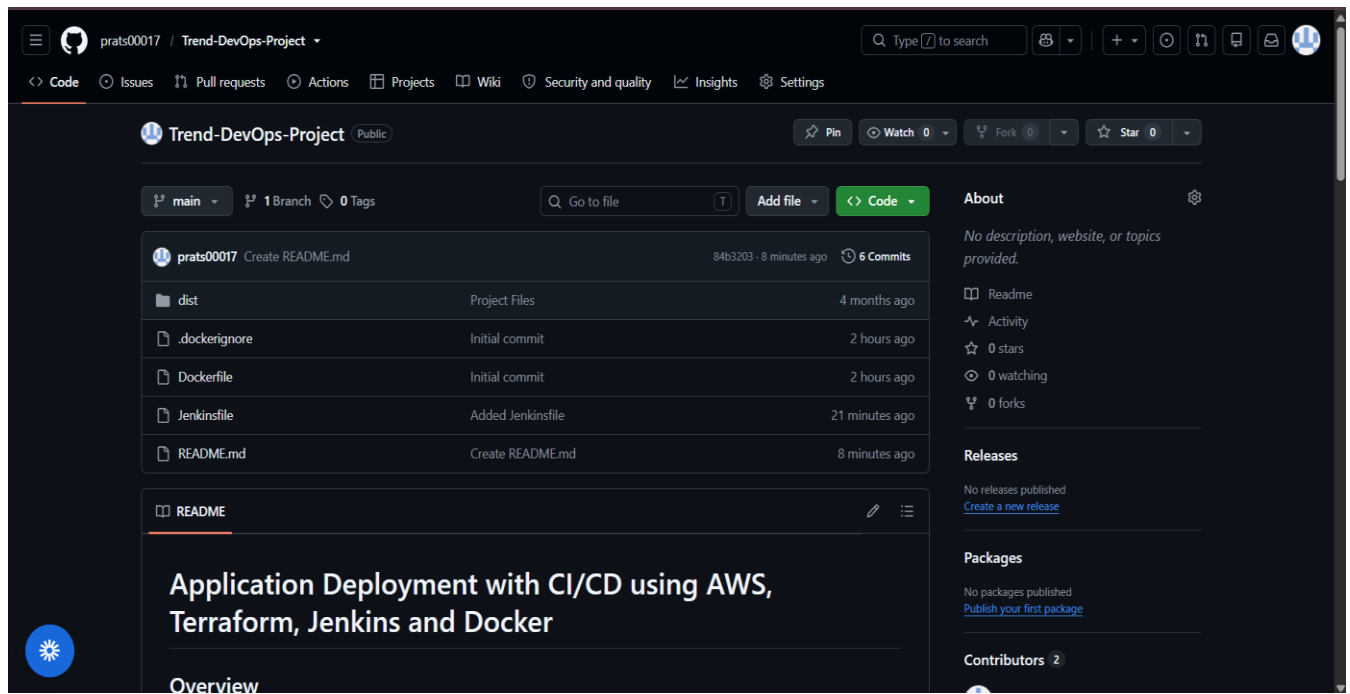
The screenshot shows the Jenkins Pipeline View for the 'Trend-DevOps-Pipeline'. The interface includes a sidebar with 'Status', 'Changes', 'Build Now', 'Configure', 'Delete Pipeline', 'Full Stage View', 'Stages', 'Rename', and 'Pipeline Syntax' links. The main area displays a 'Stage View' table with columns for 'Declarative: Checkout SCM', 'Build Docker Image', and 'Run Container'. The table shows two stages: #2 (successful) and #1 (failed). The 'Permalinks' section lists links to the last build, last stable build, last successful build, last failed build, last unsuccessful build, and last completed build.

Stage	Declarative: Checkout SCM	Build Docker Image	Run Container
#2	1s	11s	1s
#1			

Permalinks

- Last build (#2), 18 min ago
- Last stable build (#2), 18 min ago
- Last successful build (#2), 18 min ago
- Last failed build (#1), 32 min ago
- Last unsuccessful build (#1), 32 min ago
- Last completed build (#2), 18 min ago

7. GitHub repository



8. Final deployed Trendify webpage

